CR-003: Medicine Price Validation and Role-Based Discount Management

1. Change Request Details

Field	Details
Change Request ID	CR-003
Title	Medicine Price Validation and Role-Based Discount Management
Priority	High
Severity	Major
Change Type	Business Logic Enforcement / Financial Security
Project	Pharmacy Management System (PMS)
Module	Sales & Billing Management
Release Target	Sprint 4 / Release 1.4
Target System	PMS Web Application (Node.js / Express, MySQL, Vanilla JavaScript, HTML5 and CSS3)
Target Environments	Development, Staging and Production
Assigned Roles	Admin, Pharmacist and Pharmacy Staff
Document Reference	CR-003-DISC-VAL | Version 1.0.0 | October 30, 2026

2. Business Requirement

The Pharmacy Management System must ensure that medicine prices and discounts used during a sale are valid and authorized. Only the Admin role may add or manage the discount percentage options available in the system. Pharmacists and Pharmacy Staff may apply an approved discount only by selecting an available option from the dropdown menu on the Sales page. The backend must independently verify the medicine selling price and discount selection before completing a sale.

3. Current Behaviour

- The Sales page currently allows staff and pharmacists to manually enter discount values in an open text or numeric field.
- There is no centralized administrator-controlled list of approved discount options.
- Medicine prices, discount percentages and final totals can be submitted from the client interface without sufficient backend verification.
- A modified browser request or direct API request could submit an unapproved discount or manipulated price and subtotal.

4. Requested Change

Introduce administrator-controlled discount options and enforce server-side medicine price and discount validation throughout the sales process.

- Admin discount management: only an authenticated Admin may add, view, edit or deactivate discount options through the administrative discount configuration interface.
- Sales page selection: the Sales page must display a dropdown populated with active, Admin-approved discount options.
- Role restriction: Pharmacists and Pharmacy Staff may only select an existing active discount from the Sales page dropdown. They must not be able to add, edit or deactivate discount options.
- Controlled defaults: the dropdown must default to 0% and must not permit free-text discount entry.
- Backend validation: the server must verify that the selected discount is active and approved, retrieve the medicine selling price from the medicines table, and calculate the sale totals itself.
- Audit records: store the selling price, discount percentage, discount amount and subtotal for each sale item in sale_items.
- Financial precision: use consistent two-decimal currency calculations and rounding.

5. Scope

5.1 In Scope

- Create the discount_options table for approved discount percentage options.

- Verify and standardize the sale_items fields: selling_price, discount_percentage, discount_amount and subtotal.
- Provide an Admin-only discount configuration interface and protected management routes.
- Replace the free-text discount input on the Sales page with a dropdown populated with active options.
- Provide live display of gross total, discount amount and final payable amount.
- Enforce backend role checks, price lookup, discount whitelist validation and transaction safety.

5.2 Out of Scope

- Customer loyalty point accrual and redemption (identified for CR-008).
- Dynamic multi-medicine basket promotions such as Buy 2 Get 1 Free (identified for CR-009).
- Multi-currency conversion; the system uses local currency (Rs.).

6. Role Permissions and User Stories

System Role	Permitted Discount Actions	Restricted Actions
Admin	Add and manage approved discount options through the Admin discount configuration interface. Select an available discount during a sale.	Must follow the same server-side price and discount validation when creating a sale.
Pharmacist	Select and apply an active approved discount using the dropdown on the Sales page.	Cannot add, edit or deactivate discount options or access Admin discount management operations.
Pharmacy Staff	Select and apply an active approved discount using the dropdown on the Sales page.	Cannot add, edit or deactivate discount options or access Admin discount management operations.

6.1 User Story 1: Admin Discount Configuration

As an Admin, I want to add and manage approved discount percentage options so that the discounts available to sales users are controlled by pharmacy management.

Acceptance criteria: Only the Admin role can use the discount management routes. Percentages must be numeric values from 0.00% to 100.00%. Duplicate percentages must be rejected. Deactivated options are excluded from future Sales page dropdowns while historical sales remain unchanged.

6.2 User Story 2: Discount Selection by Pharmacist and Pharmacy Staff

As a Pharmacist or Pharmacy Staff member, I want to select an approved discount from the Sales page dropdown so that I can apply only discounts authorized by the Admin.

Acceptance criteria: The dropdown contains active options returned by GET /api/sales/discounts. The default is 0%. Pharmacists and Pharmacy Staff cannot create, modify or deactivate options. Selecting a discount updates the discount amount and subtotal immediately.

6.3 User Story 3: Server-Side Price and Discount Validation

As the system, I must calculate sale values using verified database prices and approved discount options so that client-side changes cannot compromise financial integrity.

Acceptance criteria: Unapproved or inactive discounts are rejected with 400 Bad Request. Client-submitted selling prices and subtotals are ignored. Server calculations are completed and validated before the sale transaction is committed.

7. Functional and Security Requirements

Requirement ID	Requirement
FR-001	Only authenticated Admin users shall add, edit, view or deactivate discount options through the Admin discount configuration interface.
FR-002	Discount percentages shall be numeric values between 0.00% and 100.00%, with duplicate percentages rejected.
FR-003	The Sales page shall retrieve and display only active, Admin-approved discount options in a dropdown.
FR-004	Pharmacists and Pharmacy Staff shall only select an existing active discount from the Sales page dropdown; they shall not manage discount options.
FR-005	The dropdown shall default to 0% and update line-item discount amount and subtotal when changed.
FR-006	The backend shall retrieve medicine selling prices from the medicines table and ignore client-submitted prices and subtotals.
FR-007	The backend shall reject unapproved, inactive or invalid discount values with a clear validation response.
FR-008	The system shall store selling_price, discount_percentage, discount_amount and subtotal for each sale item.
SEC-001	Existing authentication and sales permissions shall remain enforced.

Requirement ID	Requirement
SEC-002	Admin discount management routes shall enforce the Admin role on the server; frontend restrictions shall not replace server-side role and discount validation.

8. Technical Specifications

8.1 Database Changes

- Create discount_options with discount_id, percentage, label, is_active, created_by, created_at and updated_at fields. The percentage value must be unique.
- Verify or add sale_items snapshot fields: selling_price DECIMAL(10,2), discount_percentage DECIMAL(5,2), discount_amount DECIMAL(10,2) and subtotal DECIMAL(10,2).
- The source specification identifies baseline active options of 0%, 5%, 10%, 15% and 20%.

8.2 API Endpoints and Access

Method / Route	Purpose	Allowed Roles
GET /api/sales/discounts	Retrieve active discount options for the Sales page dropdown.	Admin, Pharmacist, Pharmacy Staff
GET /api/admin/discounts	Retrieve all discount options, including inactive options.	Admin only
POST /api/admin/discounts	Add an approved discount percentage option.	Admin only
PUT /api/admin/discounts/:id/status	Activate or deactivate a discount option.	Admin only
POST /api/sales	Create a sale using medicine identifiers, quantities and selected approved discount percentages.	Admin, Pharmacist, Pharmacy Staff

The sales endpoint must validate every item before committing sale and stock changes. Invalid or inactive discount values return 400 Bad Request. Unauthenticated or unauthorized requests return 401 Unauthorized or 403 Forbidden, as applicable.

8.3 Calculation Rules

For each item: Gross Total = database selling price x quantity. Discount Amount = rounded to two decimal places (Gross Total x selected discount percentage / 100). Subtotal = Gross Total - Discount Amount.

For the sale: Gross Sale Total = sum of item gross totals. Total Discount = sum of item discount amounts. Final Payable Amount = Gross Sale Total - Total Discount.

9. Acceptance Criteria

- AC-001: Only Admin users can add or manage discount options.
- AC-002: Pharmacists and Pharmacy Staff can view active discount options and select them from the Sales page dropdown, but cannot manage the options.
- AC-003: The Sales page dropdown contains only active approved discount options and defaults to 0%.
- AC-004: Changing the dropdown updates discount and subtotal calculations correctly.
- AC-005: Unapproved, inactive, duplicate or out-of-range discount values are rejected as specified.
- AC-006: The backend uses current database medicine prices and does not trust client-submitted price calculations.
- AC-007: Successful sale items retain the required price and discount snapshot fields.
- AC-008: Existing authentication, authorization, stock handling and normal sales workflow continue to operate.

10. Expected / Potentially Affected Files

- routes/sales.js
- routes/admin/discounts.js
- public/pages/sales.html, public/js/sales.js and public/css/sales.css
- public/pages/admin/discounts.html and public/js/admin/discounts.js
- Database schema definition (schema.sql), sale_items and discount_options.

Additional files, if modified, should be documented and justified in the Implementation Report.

11. Sign-off

Stakeholder Role	Name / Title	Signature	Date
Technical Lead	Lead Full-Stack Architect		October 30, 2026

Stakeholder Role	Name / Title	Signature	Date
Product Owner	Pharmacy Systems Product Manager		October 30, 2026
QA Lead	Quality Assurance Manager		October 30, 2026
Security & Compliance	Data Protection & Compliance Officer		October 30, 2026